

- 4 (a) Define *electromotive force* of a cell, and explain its difference from the *terminal potential difference* across the terminals of the cell.

[2]

The electromotive force of a cell is the energy converted from other forms of energy into electrical energy per unit charge driven round a complete circuit. B1

Terminal potential difference across the battery however is the e.m.f of the cell less the loss volts across the internal resistance of the cell. B1

- (b) The variation with current I of terminal potential difference V across a battery with e.m.f. E and internal resistance r is shown in Fig. 4.1.

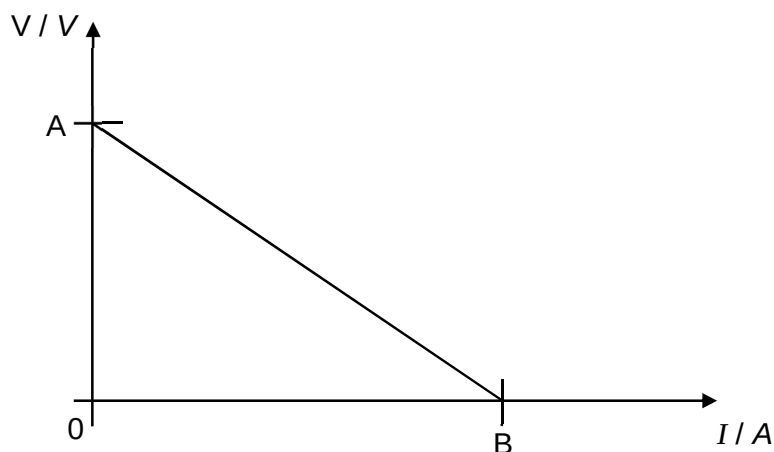


Fig. 4.1

In Fig. 4.1, state and explain the significance of the values of

- (i) the vertical intercept, A ;

[2]

The vertical intercept is the value of the e.m.f. of the battery. M1

This is the potential difference across the battery when there is no current flowing through it hence zero potential difference across the internal resistance. A1

- (ii) the horizontal intercept, B .

The horizontal intercept is the value of the maximum current that will flow through the battery when there is no (external) resistance applied across it. (i.e. short circuit current)

[2]

M1

A1

- (c) In an attempt to obtain the graph of Fig. 4.1 for the battery, a student sets up a circuit as shown in Fig. 4.2.

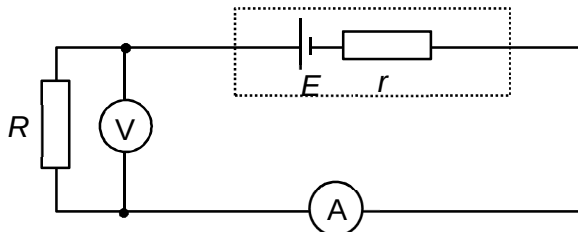


Fig. 4.2

State and explain why the circuit shown in Fig. 4.2 is inappropriate for obtaining data for the full range of values in Fig. 4.1.

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The resistor R used is a fixed resistor.

[4]

B1

This would not allow the current to vary in Fig. 4.1, but the graph in Fig. 4.1 shows that there is a varying current in the circuit. Thus it is not possible to have the variation of potential difference plotted while current is varying.

B1

In addition, the voltmeter will not be able to measure the e.m.f. of the battery as well as the current at lower values / voltage at higher values even when the fixed resistor has been replaced by a rheostat/variable resistor.

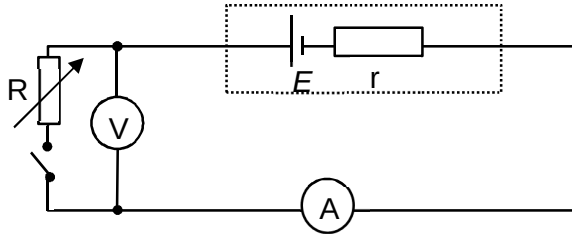
B1

This is because the rheostat cannot be adjusted to infinite resistance and thus there would always be some current flowing through the circuit.

B1

- (d) In the space below draw a circuit diagram, using the appropriate components, from which the full range of values in Fig. 4.1 may be obtained.

Solution:



- **Replace the fixed resistor with a rheostat.**
- **Add in a switch in series with the rheostat.**

B1

B1